

Pulses Insights — data flow

Camera → Pulse → Backend API → Dashboard

WRITE PATH — one message per detection

READ PATH — aggregated on request



Derived metrics — calculated in step 4, never stored

Visitors	COUNT(DISTINCT face_id) — one person counted once, however often they were seen
Visits	LAG(detected_at) per face_id — a gap longer than the threshold starts a new visit
Journeys	pulses per visit ordered by time, joined to zone, consecutive repeats collapsed
Demographics	age and gender resolved per face_id first, then aggregated — never per detection
Happiness index	emotion mapped to a score, averaged over the selected window and zone

Nothing downstream of step 3 writes to the database. Every figure above is recomputed from raw pulses, so changing a rule never needs a backfill.